

BETTINA GEORGE

gobetti@unc.edu | (919) 923-5992 | [linkedin.com/bettina-george](https://www.linkedin.com/bettina-george) | github.com/BettinaGeorge

RELEVANT EXPERIENCE

Lowe's Tech Hub | Software Engineering Intern 2X

May 2025 - Present

- Engineered a reactive, event-driven **Java/Spring Boot** quoting service integrating **MongoDB, Kafka, and internal product-data systems**, achieving **24/24 passing tests**, ~95% requirements coverage, and validation against production-stage data.
- Designed an AI-native, **specification-driven** workflow that integrated **MCP, Jira, Confluence, & AI coding agents** to convert product requirements into specifications, architecture decisions, and production code, reducing implementation rework by 3–4×.
- Rebuilt a **React/TypeScript/FastAPI** internal platform supporting **1.5K+** employees, improving office navigation and onboarding through iterative **frontend/backend** enhancements.
- Maintained **95%+** Cypress test coverage, supporting **CI/CD pipelines** by validating production features & **PostgreSQL** data integrity with **DBeaver**.

CodePath | AI Engineering Fellow

May 2026 - Present

- Built **RAG systems, multi-tool agents, content classifiers, & moderation APIs** using **Python, LangChain, ChromaDB, FastAPI, & LLM APIs**.
- Developed AI workflows with vector search, evaluation pipelines, and prompt injection defenses for reliable **LLM-powered systems**.
- Contributed to **production-grade open-source software** through pull requests, code reviews, Git workflows, and collaborative development.

KPMG + Break Through Tech | AI/ML Fellow

May 2025 - Jun. 2026

- Built and evaluated supervised machine learning models in **Python (Pandas, scikit-learn)** on real-world productivity and energy datasets.
- Developed an **optimization system (NSGA-II)** to **quantify trade-offs** between generative AI productivity gains and energy consumption.
- Analyzed model outputs and **translated metrics** into actionable insights for stakeholders to support responsible, scalable AI deployment.

UNC Department of Computer Science | Undergraduate Teaching Assistant (COMP 110)

Aug 2025 - May 2026

- Selected from 180+ applicants to mentor students in **Python, debugging, & software development** in an introductory programming course.
- Led weekly labs/office hours as a technical point of contact, diagnosing issues and communicating complex concepts under time constraints.

PROJECTS

ContextOS | AI Developer Platform

[Project Link](#)

- Built an **AI-native** platform connecting **GitHub & Notion** to help developers understand codebases through evidence-backed, source-cited retrieval.
- Engineered **RAG** pipeline using **PostgreSQL/pgvector, Voyage embeddings, & LLM APIs** with citations, prompt-injection defense, and abstention.
- Architected a modular connector framework with **FastAPI, Next.js, & Supabase** to support integrations with **Jira, Figma, Slack, & engineering tools**.

Quiet Table | Pearl Hacks Hackathon 2026

[Project Link](#)

- Won **1st Place at Pearl Hacks 2026** for building a privacy-first, real-time dining connection app with ephemeral, no-profile interactions.
- Implemented geospatial proximity matching with **MongoDB 2dsphere indexes and Google Maps API** for real-time location discovery.
- Led **frontend** development in **Next.js/TypeScript** and collaborated on **Node.js/Express APIs** for real-time flows and **Gemini-powered** icebreakers

CreatorOS | Full-Stack AI

[Project Link](#)

- Built a **full-stack** platform using **FastAPI, Next.js/TypeScript, and PostgreSQL** to analyze Instagram Reel performance and guide content strategy.
- Engineered a pluggable **ingestion layer** parsing **Instagram Reel JSON** exports into **PostgreSQL**, enabling a swap to the **Instagram Graph API**.
- Integrated **Anthropic Claude** to power **AI creator tools** (hooks, briefs, strategy insights, trend scouting) using live analytics context;

De-Impact Web App | JP Morgan Code for Good Hackathon 2024

[Project Link](#)

- Collaborated with 6 others to create a **web app** that connects civilians with NGOs through a skill/interest-based matching system.
- Designed a seamless, **user-centered interface** using **Figma** and implemented the front end using **React.js, TypeScript, and Tailwind CSS**.
- Assisted in building a **60% matching compatibility algorithm** with **Java/Spring Boot/MySQL**, boosting demo engagement by 25%.

Diaspora Duo App | Pearl Hacks Hackathon 2024

[Project Link](#)

- Built an AI-powered web app using **Flask & OpenAI APIs** to deliver personalized, multilingual guidance for immigrants navigating a new country.
- Awarded **Best DEI Hack** by **Fidelity** and **Best Use of AI** by **Infosys** among 70+ competing teams.

LEADERSHIP & ACTIVITIES

- Black in Technology:** Project Development Chair; **led technical team and event planning**; increased student engagement by 32%.
- SEO Career Prep:** Selected for 100+ hours of career coaching and software engineering training.
- African Student Association:** Events Planning Committee Member & First Year Liaison.
- Other Memberships:** ColorStack, Rewrite The Code, Codepath, UNC CS+Social Good, AppTeam Carolina, Kappa Theta Pi

EDUCATION

University of North Carolina at Chapel Hill | Dean's List

Graduating May 2027

Bachelor of Science in Computer Science & Information Science

GPA: 3.79

Relevant Coursework: DSA, OOP, Algorithms & Analysis, Intro to Cybersecurity, Modern Web, Computer Organization, DB Concepts/Apps, Info Systems Analysis & Design, AI/ML Foundations, Foundations of Software Engineering, Applications of AI Engineering

SKILLS

Languages: Python, Java, C, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks/Libraries: React, Next.js, Node.js, Spring Boot, Flask, Kafka, Pandas, NumPy, Matplotlib

AI Engineering/Other Tools: RAG, LangChain, ChromaDB, LLM, APIs, PostgreSQL, MongoDB, MySQL, Supabase, Drizzle ORM, Docker, Git/GitHub, GitHub Spec Kit, Devin, AWS, Microsoft Azure, DBeaver, Jira